Role of Mathematics in Data Science and How to learn it?

What is Data Science?

Data Science is a systematic process of:

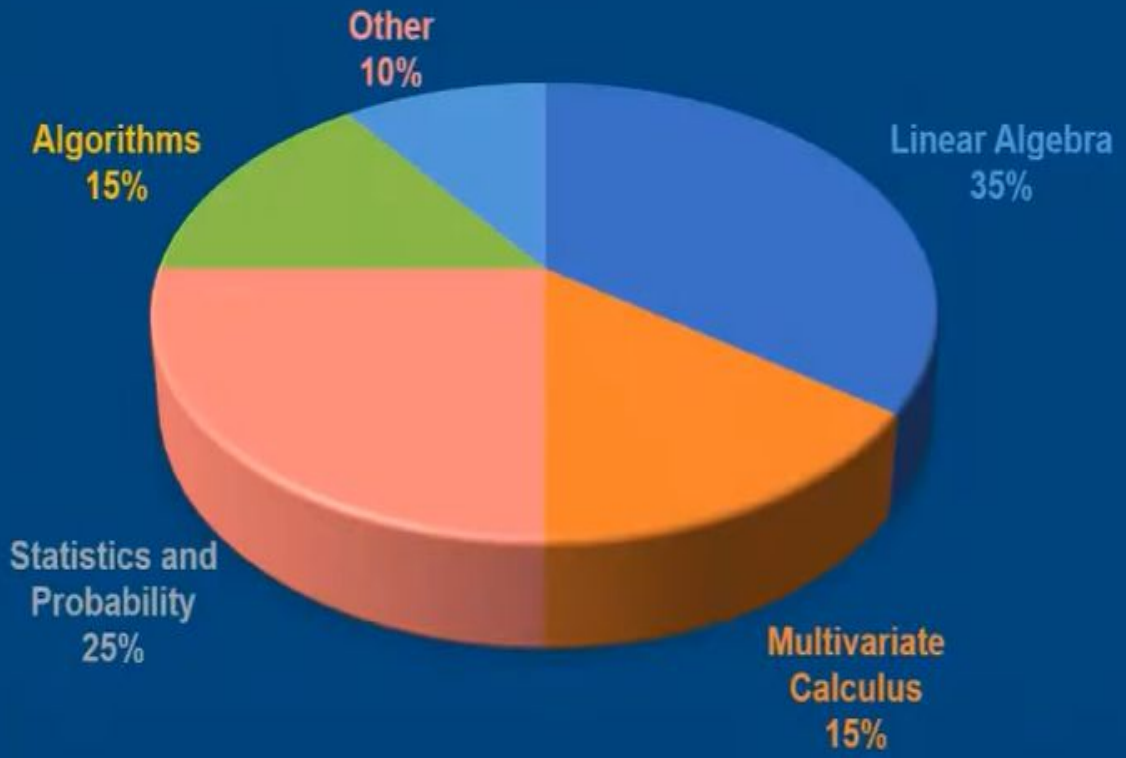
1. Collecting data
2. Representing data numerically
3. Finding patterns
4. Making decisions based on evidence

Key Idea:

> Computers do not understand emotions or language — they understand **numbers**.

This is why mathematics is essential.

Why Mathematics in Machine Learning?



This picture is a **pie chart** that explains the importance of different branches of **mathematics** in **Machine Learning (ML)**. It shows how much each area of mathematics contributes to ML.

Linear Algebra – 35% (Most Important)

- It is the **foundation of Machine Learning**
- Used for vectors, matrices, and matrix operations
- Helps represent data in numerical form
  Example: Images, datasets, neural network weights

Statistics and Probability – 25%

- Used to **analyze data** and make predictions
- Helps measure uncertainty, accuracy, and error
  Example: Mean, variance, probability, model evaluation

Multivariate Calculus – 15%

- Used to **train machine learning models**
- Helps minimize error using derivatives
  Example: Gradient Descent, loss minimization

Algorithms – 15%

- Provide step-by-step procedures
- Define how machine learning models work
  Example: KNN, Decision Trees, Support Vector Machines

Other – 10%

- Includes optimization techniques, numerical methods, and other math tools

1. What is a Scalar?

A **scalar** is a **single numerical value**.

It represents **only magnitude (quantity)** and has **no direction**.

In simple words:

Scalar = one number that tells how much

2. Real-Life Examples of Scalars

- Age = **20**
- Salary = **50,000**
- Temperature = **35°C**
- Weight = **70 kg**
- Speed = **60 km/h**
- Accuracy = **0.92**

All these values are **scalars** because they consist of **only one number**.

3. Scalars in Mathematics

In mathematics and linear algebra:

- Scalars are usually represented by **small letters** or **Greek symbols** (a, b, c, α, λ)

a = 5

b = -3

α = 0.01

Scalars can be:

- **Positive**
- **Negative**
- **Zero**
- **Decimal values**

4. Scalar Operations

4.1 Scalar Multiplication

A scalar can multiply a vector or a matrix.

Example:

Scalar = 2

Vector = [1, 2, 3]

Result = 2 × [1, 2, 3]

= [2, 4, 6]

This operation is called **scalar multiplication** and it **scales** the vector.

4. Scalar Operations

Scalar in Linear Algebra

In linear algebra:

- Scalars control the **size or strength** of vectors and matrices
- They are used in:
  - Vector scaling
  - Matrix scaling
  - Linear transformations

Without scalars, vectors and matrices cannot be adjusted or tuned.

4. Scalar Operations

Scalars control how much change happens

Vectors and matrices **hold the data**, but **scalars decide the intensity**

Volume Knob Example

- **Song** → Vector (many sound frequencies)
- **Playlist** → Matrix (many songs)
- **Volume knob** → Scalar

Question:
Can you make the song louder or softer **without** the volume knob?

No.

Same song (vector), same playlist (matrix)
Only **scalar** changes how strong the sound is

Scalar in Programming (Python Example)

In Python, a scalar is stored in a simple variable.

Example:

```
learning_rate = 0.01

loss = 2.4

accuracy = 0.93
```

Using NumPy:

```
import numpy as np

x = np.array(5)
```

Properties of Scalars

Scalars follow standard arithmetic rules.

11.1 Closure

If you add, subtract, multiply, or divide two scalars, the result is also a scalar.

Example:

```
5 + 3 = 8

10 ÷ 2 = 5
```

Properties of Scalars

11.2 Commutative Property

Order does not matter.

$$a + b = b + a$$

$$a \times b = b \times a$$

11.3 Associative Property

Grouping does not matter.

$$(a + b) + c = a + (b + c)$$

Properties of Scalars

11.4 Distributive Property

Scalar distributes over vectors.

```
a(b + c) = ab + ac
```

Scalars in Machine Learning (VERY important)

(a) Model parameters

Many ML parameters are scalars:

- Learning rate = `0.01`
- Regularization strength (λ) = `0.1`
- Bias term = scalar

Example (linear regression):

$$y = wx + b$$

- x = scalar input
- w = scalar weight
- b = scalar bias
- y = scalar output

Scalars in Machine Learning (VERY important)

ML Example (Student Performance)

Suppose:

- Attendance weight = **0.4**
- CGPA weight = **0.6**

Student:

- Attendance = 80
- CGPA = 3.5

Score:

$(0.4\times80)+(0.6\times3.5) = 34.1$

Each multiplication uses **scalars**

Identify Scalars in the Equation

$$y = 2.5x + 1.8$$

Identify Scalars in the Equation

$$y = 2.5x + 1.8$$

Scalars are:

- **2.5** → weight (scalar)
- **1.8** → bias (scalar)

Explanation:

- **x** is a variable (input feature)
- **y** is output
- **2.5** and **1.8** are **fixed single numbers**
  Single number = Scalar

Answer:
Scalars = 2.5 and 1.8

1. What is a Vector? (Core Idea)

A **vector** is:

An ordered collection of numbers that has **magnitude and direction**
OR (in data science terms)
An **ordered list of scalars** representing features or values.

Mathematical notation

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

Example:

$$\vec{v} = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$$

Why Vectors Matter in Data Science

Data science is about **analyzing real-world data** using computers.
But computers **cannot understand real-world objects directly** such as:

- people
- text
- images
- sounds

So the **first step in data science** is:

Convert real-world data into **numbers**

These numbers are stored and processed in the form of **vectors**.

Vectors are the foundation of data science and machine learning.

What a Vector Means in Real Data Science

A structured list of numerical features that describes **one real-world entity** so it can be analyzed by a computer.

Examples of real-world entities:

- a customer
- a student
- a product
- a document
- an image

Each becomes **one vector**.

In data science, a **vector** is:

A **one-dimensional ordered collection of numbers** that represents data.

Example:

```
[25, 60000, 3]
```

This vector might represent:

- Age = 25
- Salary = 60000
- Experience = 3 years

Each number is a **feature**.

What is a Vector? (Core Idea)

3. Types of Vectors

(a) Row Vector

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

(b) Column Vector

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Simple real-life example

Age	Marks	Attendance
20	85	90

This student is represented as a **vector**:

[20, 85, 90]

One row = one vector

Dataset structure

Student	Vector
A	[20, 85, 90]
B	[22, 78, 80]
C	[19, 92, 95]

- Each **row** is a vector
- The full table is a **matrix**

Text data example (NLP)

Sentence:

```
"I love data science"
```

Computer converts it into numbers:

```
[0.12, 0.45, 0.78, 0.33]
```

Text becomes a vector

Used in:

- chatbots
- search engines
- recommendation systems

Image data example

An image is made of pixels.

Computer sees an image as:

```
[255, 240, 120, 60, ...]
```

✔ **Image = vector**

Similarity using vectors (Easy idea)

Two students:

```
A = [20, 85, 90]

B = [21, 83, 88]
```

These vectors are **close**, so students are **similar**.

Another student:

```
C = [30, 40, 50]
```

This vector is **far**, so student is **different**.

Real-World Example 1: Customer Data (E-commerce)

Suppose an online store tracks this data:

Feature	Value
Age	28
Monthly Spend	45,000
Visits per Month	12

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Suppose an online store tracks this data:

Feature	Value
Age	28
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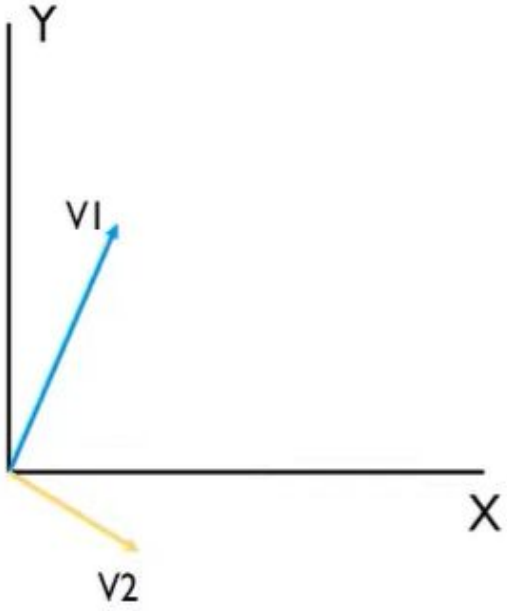
Customer vector:

```
[28, 45000, 12]
```

One customer = one vector
Many customers = many vectors (dataset)



Vector Addition (Dot Product)

- Adding 2 Vectors is not simple Arithmetic
- It is the total work achieved through both vectors in a quantified form

Student Performance Analysis Using Vectors

1. Problem Overview

A university wants to build an **intelligent academic support system** that can:

1. Analyze student performance
2. Identify weak and strong students
3. Group similar students
4. Recommend extra classes or scholarships

Since computers do not understand humans directly, **student information must be converted into numerical form**.

2. Defining Student Features

The university decides to use the following **three features** for each student:

- **Age (years)**
- **CGPA (out of 4.0)**
- **Attendance percentage (%)**

Each student will be represented as a **feature vector**.

3. Representing Students as Vectors

Given Student Data

Student	Age	CGPA	Attendance
Student A	20	3.2	85
Student B	22	3.8	92
Student C	21	3.4	88

3. Representing Students as Vectors

Vector Representation

$$\mathbf{A} = [20,\ 3.2,\ 85]$$

$$\mathbf{B} = [22,\ 3.8,\ 92]$$

$$\mathbf{C} = [21,\ 3.4,\ 88]$$

Each student is now represented as a **3-dimensional vector**, where:

- Each element is one feature
- One student = one vector
- All students together form a **dataset**

Using Rules to Classify Students

Before ML, a **rule-based approach** can be used:

Example rules:

1. **If CGPA < 3.0 → student is weak**
2. **If Attendance < 75% → student is weak**
3. **Otherwise → student is strong**

Let's see our students:

- **A = [20, 3.2, 85] → CGPA > 3.0, Attendance > 75 → Strong**
- **B = [22, 3.8, 92] → Strong**
- **C = [21, 3.4, 88] → Strong**

If we had:

- **D = [20, 2.8, 80] → CGPA < 3.0 → Weak**
- **E = [21, 3.1, 70] → Attendance < 75 → Weak**

This is very simple but works for small datasets.

Vector Representation

$$\mathbf{A} = [20,\ 3.2,\ 85]$$

$$\mathbf{B} = [22,\ 3.8,\ 92]$$

$$\mathbf{C} = [21,\ 3.4,\ 88]$$

4. Why Feature Scaling Is Necessary

Problem Without Scaling

- Age values are around 20
- CGPA values are around 3
- Attendance values are around 90

Attendance has much larger numerical values and may **dominate the learning process**, causing biased results.

5. Feature Scaling (Simple Normalization)

To bring all features to a **similar range**, we scale them.

Scaling Formula (Simple Approach)

$$\text{Scaled value} = \frac{\text{Actual value}}{\text{Maximum possible value}}$$

Scaled Student A Vector

$$\mathbf{A}_{scaled} = \left[\frac{20}{25},\ \frac{3.2}{4},\ \frac{85}{100}\right] = [0.8,\ 0.8,\ 0.85]$$

Explanation

Now:

- All features are between 0 and 1
- No feature dominates another
- Machine Learning models learn more efficiently

Identify Weak and Strong Students

Problem: Among hundreds of students, who needs help?

Solution: Feature Scaling + Thresholds

Scale features so that CGPA, Attendance, Age are comparable.

```
Scaled A = [0.8, 0.8, 0.85]

Scaled B = [0.88, 0.95, 0.92]

Scaled C = [0.84, 0.85, 0.88]
```

Define a **performance score** using weighted sum or ML:

```
Performance_score = 0.6*CGPA + 0.4*Attendance
```

Compare scores:

```
A = 0.6*0.8 + 0.4*0.85 = 0.82

B = 0.6*0.95 + 0.4*0.92 = 0.94

C = 0.6*0.85 + 0.4*0.88 = 0.86
```

- **B** → strong, **A** → needs improvement
- **Practical ML:** Decision Tree can automatically **classify students** into strong/weak groups

6. Measuring Similarity Using Euclidean Distance

The university wants to know:

> Which student is most similar to **Student A**?

Euclidean Distance Formula

$$d = \sqrt{\sum (x_i - y_i)^2}$$

Distance Between Student A and B

$$\begin{aligned} d(A, B) &= \sqrt{(22-20)^2 + (3.8-3.2)^2 + (92-85)^2} \\ &= \sqrt{4 + 0.36 + 49} = \sqrt{53.36} \approx 7.3 \end{aligned}$$

Distance Between Student A and C

$$\begin{aligned} d(A, C) &= \sqrt{(21-20)^2 + (3.4-3.2)^2 + (88-85)^2} \\ &= \sqrt{1 + 0.04 + 9} = \sqrt{10.04} \approx 3.16 \end{aligned}$$

Interpretation

- Distance A–C is smaller than A–B
- **Student C is more similar to Student A**

7. Using Dot Product for Academic Strength Alignment

The university also wants to measure **overall academic alignment**.

Dot Product Formula

$$\mathbf{A} \cdot \mathbf{B} = \sum (a_i \times b_i)$$

Dot Product of Student A and B

$$(20 \times 22) + (3.2 \times 3.8) + (85 \times 92)$$

$$= 440 + 12.16 + 7820 = 8272.16$$

Interpretation

- A higher dot product means stronger alignment
- Used for **ranking**, **recommendations**, and **performance comparison**

1. Compare dot products

Example:

A = [20, 3.2, 85]
B = [22, 3.8, 92] → dot product = 8272.16
C = [18, 2.5, 60] → dot product with A = ?

Compare dot products

Example:

A = [20, 3.2, 85]
B = [22, 3.8, 92] → dot product = 8272.16
C = [18, 2.5, 60] → dot product with A = ?

(20*18) + (3.2*2.5) + (85*60)
= 360 + 8 + 5100
= 5468

Interpretation:

- **A · B = 8272.16**
- **A · C = 5468**

Conclusion: **Student B is more aligned with Student A than Student C**, because 8272 > 5468.

Step 1: Define Ideal Student

```
Ideal = [22, 4.0, 95]
```

Step 2: Compute Dot Product with Real Students

```
Student A = [20, 3.2, 85]
Dot(A, Ideal) = 440 + 12.8 + 8075 = 8527.8

Student B = [22, 3.8, 92]
Dot(B, Ideal) = 484 + 15.2 + 8740 = 9239.2
```

Step 3: Set Threshold

```
Threshold = 9000
```

Step 4: Decide

- A: 8527.8 < 9000 → **No Scholarship**
- B: 9239.2 > 9000 → **Eligible for Scholarship**

8. Normalization (Direction-Based Comparison)

What is Normalization?

Normalization means converting a vector to **unit length** (magnitude = 1) **without changing its direction**.

$$\mathbf{v}_{norm} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

1. Imagine Walking to School

- Suppose you are walking from **home to school**.

- Your **path** can be represented as a **vector**.
  - **Direction** → which way you are walking (towards the school)
  - **Magnitude** → how far you are walking

2. Example Vector

- Let's say your displacement vector is:

$$\mathbf{v} = [3 \text{ km East}, 4 \text{ km North}]$$

- Magnitude (distance) =

$$||\mathbf{v}|| = \sqrt{3^2 + 4^2} = 5 \text{ km}$$

- Normalize the vector:

$$\hat{\mathbf{v}} = [3/5, 4/5] = [0.6, 0.8]$$

Same direction → still pointing **towards school**

Magnitude = 1 → now it's a **unit vector**

Example

2. Map/Navigation Example

- GPS vector: `[dx, dy]` = 10 meters East, 20 meters North
- Normalize → `[0.447, 0.894]` → still same **direction (towards destination)**, but length = 1 (unit vector)
- Useful for **direction-only calculations** like "which way to point the arrow"

Why Do We Need Normalization?

- In many ML problems, **magnitude of features can differ a lot**.
- Example:

```
Student A = [20, 3.2, 85]   # Age, CGPA, Attendance

Student B = [22, 3.8, 92]
```

- Attendance values (85, 92) are **much larger than CGPA** (3.2, 3.8)
- If we **compare vectors directly**, features with bigger numbers dominate the comparison.

Normalization fixes this, so we only care about **direction (pattern of features)**, not absolute scale.

Normalize the Student A and Student B by using the below formula

$$\mathbf{v}_{norm} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

A = [20, 3.2, 85]
B = [22, 3.8, 92]

A = [20, 3.2, 85]
B = [22, 3.8, 92]

Step 2: Find the magnitude of each vector

Student A:

$$|A| = \sqrt{20^2 + 3.2^2 + 85^2} = \sqrt{400 + 10.24 + 7225} = \sqrt{7635.24} \approx 87.38$$

Student B:

$$|B| = \sqrt{22^2 + 3.8^2 + 92^2} = \sqrt{484 + 14.44 + 8464} = \sqrt{8962.44} \approx 94.66$$

Step 3: Divide each component by magnitude (Normalization)

Normalized A:

$$A_{norm} = \left[\frac{20}{87.38}, \frac{3.2}{87.38}, \frac{85}{87.38}\right] \approx [0.229, 0.037, 0.973]$$

Normalized B:

$$B_{norm} = \left[\frac{22}{94.66}, \frac{3.8}{94.66}, \frac{92}{94.66}\right] \approx [0.232, 0.040, 0.972]$$

9. Final Decision-Making by the System

Based on vector analysis:

Condition	Decision
Low attendance	Warning issued
Close distance to weak students	Extra coaching
High dot product	Scholarship consideration
High similarity	Same study group

Student Performance Analysis Using Vectors

Instructions: Solve the following problems step by step. Show all calculations clearly.

Problem 1: Represent Student as Vector

A student has the following details:

- Age = 21 years
- CGPA = 3.5
- Attendance = 88%

Tasks:

1. Represent this student as a **vector**.
2. Write the **dimension** of the vector.

Problem 2: Represent Multiple Students

Three students have the following data:

Student	Age	CGPA	Attendance
A	20	3.2	85
B	22	3.8	92
C	21	3.4	88

Tasks:

1. Represent each student as a **vector**.
2. Identify the **strongest and weakest student** based on CGPA and Attendance only.

Problem 3: Feature Scaling

- Scale the features of **Student A** from Problem 2 to **0–1 range**.
- Use **max value method**:

$$x_{scaled} = \frac{x}{x_{max}}$$

Problem 4: Dot Product for Academic Alignment

- Take **Student A** and **Student B** from Problem 2.
- Compute the **dot product**.
- Explain what a **higher dot product** means in terms of **student performance alignment**.
- Decide which student is more likely to get a **scholarship** if ideal student vector = `[22, 4, 95]`.

Problem 5: Euclidean Distance

- Compute the **Euclidean distance** between **Student A** and **Student C**.
- Explain how **smaller distance** can be used to **group similar students**.

Problem 6: Normalization

- Normalize **Student A** and **Student B** vectors.
- Show all calculations.
- Explain **why normalization is important** when comparing students' performance **patterns**.

Problem 7: Real-World Application

- Imagine a university wants to:

    1. **Identify weak students** → Extra classes
    2. **Identify strong students** → Scholarships
    3. **Group similar students** → Study groups

Tasks:

1. Use **dot product**, **distance**, or **normalized vectors** to explain **how the system decides**.
2. Write a **short paragraph** describing the steps in your own words.

Percentages, Proportions, Ratios & Growth Rates

Percentages

Introduction: Why Percentages Matter in Data Science

- Percentages are everywhere in **data analysis, ML, and statistics**.
- Common uses:
  - Data cleaning: % missing values, % duplicates.
  - Data exploration: % of classes in classification.
  - Model evaluation: Accuracy, precision, recall (%).
  - Business KPIs: Revenue growth %, churn rate %, engagement %.

Example:
If a dataset has 10,000 customer reviews, and 2,500 are positive, then the positive reviews = **25% of total reviews**.

2 Basic Concept of Percentages

- **Definition:** Percentage means "per hundred".

$$\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100$$

Example in Data Science:

- Dataset has 12,000 rows. 3,000 rows have missing values.

$$\text{Percentage of missing values} = \frac{3000}{12000} \times 100 = 25\%$$

- **Tip:** Percentages help compare features on the same scale.

Percentages in Classification & ML

- In **classification datasets**, class imbalance is measured with percentages.
- **Example:**
  - Dataset has 5,000 samples.
  - Class A: 4,000 (80%), Class B: 1,000 (20%)
- **Why important?** Models trained on imbalanced datasets may **favor the majority class**.

Exercise idea:

- If Class C is added with 500 samples, recalculate class percentages.

4 Percentage Increase & Decrease

- **Increase:**

$$\text{Percentage Increase} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \times 100$$

- **Decrease:**

$$\text{Percentage Decrease} = \frac{\text{Old Value} - \text{New Value}}{\text{Old Value}} \times 100$$

Example in Data Science:

- Users on a website: 50,000 → 65,000

$$\text{Increase} = \frac{65000 - 50000}{50000} \times 100 = 30\%$$

- **Example in ML:** Model accuracy improved from 75% → 87%

$$\text{Increase in accuracy} = \frac{87 - 75}{75} \times 100 = 16\%$$

Percentages for Missing Values & Data Cleaning

- **Data Cleaning:** % of missing values tells you **how much data to drop or impute**.

Example:

- Dataset: 20,000 rows, Feature X has 3,000 missing values.

  Decision: If >50% missing → drop column; else → impute.

Percentages in Feature Engineering

- **Feature ratios:** Converting counts to percentages makes features comparable.
- Example:
  - Column “Number of Purchases” = [2, 5, 10]
  - Convert to % of total purchases = `[2/17*100, 5/17*100, 10/17*100] ≈ [11.8%, 29.4%, 58.8%]`
- Helps ML models **treat features equally**.

•

7 Percentages in Growth Rates

- Often, data changes over time → % growth is used.

$$\text{Growth Rate (\%)} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \times 100$$

Example:

- Dataset of monthly active users: 50,000 → 65,000 → Growth = 30%
- **Compound growth:**

$$\text{Final} = \text{Initial} \times (1 + r_1) \times (1 + r_2) \ldots$$

- Example: Stock price grows 10% first year, 15% second year:

$$\text{Final} = P \times 1.10 \times 1.15$$

Ratios and Proportions in Data Science

1. Introduction

In Data Science, **ratios and proportions** help us:

- Compare features across data points
- Normalize data
- Compute probabilities and percentages
- Analyze trends

Example:

- In a dataset of students: ratio of males to females
- In sales data: proportion of product sales in different regions

2. Ratios in Data Science

Definition:
A **ratio** is a way to compare two quantities.

Example 1 – Feature Comparison:
Suppose we have two students:

Student	Math Marks	Science Marks
A	80	60
B	70	90

- Ratio of Math to Science marks for A:

$$80 : 60 = 4 : 3$$

- Ratio of Math to Science marks for B:

$$70 : 90 = 7 : 9$$

Interpretation:

- A performs better in Math compared to Science.
- B performs better in Science compared to Math.

Example 2 – Dataset Distribution:

If a dataset has 300 males and 200 females:

$$\text{Ratio (Male : Female)} = 300 : 200 = 3 : 2$$

This helps in **class imbalance analysis** in machine learning.

3. Proportions in Data Science

Definition:

A **proportion** states that **two ratios are equal**.

Applications:

1. Feature scaling
2. Probability calculations
3. Sampling data

Example 1 – Feature Scaling:

Suppose a dataset column `Age` ranges from 20 to 60. We want to scale it so that max = 1.

$$\frac{\text{Age} - \text{Min}}{\text{Max} - \text{Min}} = \text{Scaled Age}$$

This is **direct proportion**: old range → new range.

Example 2 – Predicting Outcomes:

Suppose 40% of users clicked an ad on mobile, and we have 250 mobile users. How many likely clicked?

$$\text{Clicked Users} = 40\% \times 250 = 100$$

This is **proportion applied to datasets.**

4. Direct and Inverse Proportions in Data Science

Direct Proportion:

- Two quantities increase or decrease together.
- Example: More hours spent on a course → more points scored.

Inverse Proportion:

- One quantity increases, other decreases.
- Example: More servers → less time to process a dataset.

Growth Rates in Data Science

1. Introduction

In Data Science, **growth rates** are used to measure how a quantity changes over time. They are essential in:

- Time-series analysis
- Trend analysis
- Forecasting
- Evaluating performance metrics

Example:

- Website traffic growth
- Stock price growth
- User adoption rate
- Sales increase over months

2. Key Concepts

a) Absolute Growth

The **absolute growth** is the **difference** between the new value and the old value.

$$\text{Absolute Growth} = \text{New Value} - \text{Old Value}$$

Example:

A website had 1,000 users in January and 1,200 users in February.

$$\text{Absolute Growth} = 1,200 - 1,000 = 200 \text{ users}$$

b) Growth Rate (%)

The **growth rate** measures the **percentage change** over time.

$$\text{Growth Rate (\%)} = \frac{\text{New Value - Old Value}}{\text{Old Value}} \times 100$$

Example:

Continuing the website example:

$$\text{Growth Rate} = \frac{1,200 - 1,000}{1,000} \times 100 = \frac{200}{1,000} \times 100 = 20\%$$

✅ The website grew by **20%** from January to February.

c) Compound Growth Rate

In many datasets, growth is **repeated over multiple periods**. We use the **Compound Annual Growth Rate (CAGR)** formula:

$$\text{CAGR} = \left(\frac{\text{Final Value}}{\text{Initial Value}}\right)^{\frac{1}{n}} - 1$$

Where:

- `Final Value` = value at the end
- `Initial Value` = value at the start
- `n` = number of periods (years, months, etc.)

Example:

A company had revenue $50,000 in 2020 and $80,000 in 2023. CAGR:

$$\text{CAGR} = \left(\frac{80,000}{50,000}\right)^{\frac{1}{3}} - 1 = (1.6)^{0.3333} - 1 \approx 1.166 - 1 = 0.166 \approx 16.6\%$$

✅ Revenue grew **16.6% per year** on average.

Percentages, Proportions, Ratios & Growth Rates

Section A: Percentages

1. The population of a town increased by 12% in 2025 and then by 8% in 2026. If the initial population was 50,000, what is the population at the end of 2026?
   (Hint: Use compound percentage growth)

2. A company reduces the price of a product by 15%, and the new price becomes $170. What was the original price?

3. If 40% of a number is equal to 30% of another number, find the ratio of the two numbers.

4. A sum of money is invested at a rate of 6% per annum compound interest. If it doubles in 12 years, what was the effective annual growth rate?

5. The price of a stock increased by 25% in the first year and decreased by 20% in the second year. If the current price is $240, what was the original price?

Percentages, Proportions, Ratios & Growth Rates

Section B: Ratios and Proportions

6. The ratio of students in Class A to Class B is 7:9. If 20 students are transferred from Class B to Class A, the ratio becomes 8:7. Find the original number of students in each class.

7. Divide $24,000 among three partners A, B, and C such that the ratio of A:B:C is 3:5:7. After one year, A receives an additional $600. What is the new ratio of A:B:C?

8. If x:y=5:7x:y = 5:7x:y=5:7 and y:z=14:15y:z = 14:15y:z=14:15, find the ratio x:y:zx:y:zx:y:z.

9. A mixture contains milk and water in the ratio 7:3. If 20 liters of water is added, the ratio becomes 7:5. Find the initial quantity of milk and water.

10. Two numbers are in the ratio 5:6. If 15 is added to each number, the ratio becomes 3:4. Find the numbers.

Percentages, Proportions, Ratios & Growth Rates

Section C: Growth Rates

11. The GDP of a country was $500 billion in 2020 and $640 billion in 2023. Calculate the **annual growth rate** (compound growth).

12. A population of bacteria doubles every 3 hours. If there are initially 500 bacteria, how many will there be after 15 hours?

13. A salary is increased by 10% every year. If the current salary is $88,000, what was the salary 3 years ago?

14. The price of a commodity increases by 12% annually for two years. If the initial price is $400, find the price after two years and the **average annual growth rate**.

Percentages, Proportions, Ratios & Growth Rates

Section D: Word/Applied Problems

16. The ratio of men to women in a company is 4:5. After hiring 20 men and 10 women, the ratio becomes 3:4. Find the original number of men and women.

17. A fund earns 8% compounded annually. How many years will it take for the investment to grow 1.5 times?

18. In a business, the profit is divided among three partners in proportion to their investments. If the total profit is $24,000 and the investments are in ratio 5:7:8, how much does each partner get?

19. A product price depreciates by 10% every year. If the current price is $729, what was the price 3 years ago?

20. The ratio of two numbers is 7:9. If 28 is added to each number, the ratio becomes 2:3. Find the numbers and calculate the percentage increase in each.